

Theorem 10.2 : *The lengths of tangents drawn from an external point to a circle are equal.*

Proof : We are given a circle with centre O, a point P lying outside the circle and two tangents PQ, PR on the circle from P (see Fig. 10.7). We are required to prove that $PQ = PR$.

For this, we join OP, OQ and OR. Then $\angle OQP$ and $\angle ORP$ are right angles, because these are angles between the radii and tangents, and according to Theorem 10.1 they are right angles. Now in right triangles OQP and ORP,

$$OQ = OR$$

(Radii of the same circle)

$$OP = OP$$

(Common)

Therefore,

$$\Delta OQP \cong \Delta ORP$$

(RHS)

This gives

$$PQ = PR$$

(CPCT)

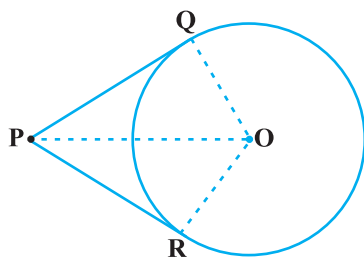


Fig. 10.7

Remarks

1. The theorem can also be proved by using the Pythagoras Theorem as follows:

$$PQ^2 = OP^2 - OQ^2 = OP^2 - OR^2 = PR^2 \text{ (As } OQ = OR\text{)}$$

which gives $PQ = PR$.

2. Note also that $\angle OPQ = \angle OPR$. Therefore, OP is the angle bisector of $\angle QPR$, i.e., the centre lies on the bisector of the angle between the two tangents.

Let us take some examples.

Example 1 : Prove that in two concentric circles, the chord of the larger circle, which touches the smaller circle, is bisected at the point of contact.

Solution : We are given two concentric circles C_1 and C_2 with centre O and a chord AB of the larger circle C_1 which touches the smaller circle C_2 at the point P (see Fig. 10.8). We need to prove that $AP = BP$.

Let us join OP. Then, AB is a tangent to C_2 at P and OP is its radius. Therefore, by Theorem 10.1,

$$OP \perp AB$$

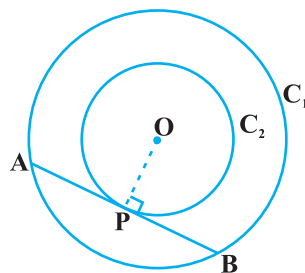


Fig. 10.8

Now AB is a chord of the circle C_1 and $OP \perp AB$. Therefore, OP is the bisector of the chord AB, as the perpendicular from the centre bisects the chord,

i.e., $AP = BP$

Example 2 : Two tangents TP and TQ are drawn to a circle with centre O from an external point T. Prove that $\angle PTQ = 2 \angle OPQ$.

Solution : We are given a circle with centre O, an external point T and two tangents TP and TQ to the circle, where P, Q are the points of contact (see Fig. 10.9). We need to prove that

$$\angle PTQ = 2 \angle OPQ$$

Let $\angle PTQ = \theta$

Now, by Theorem 10.2, $TP = TQ$. So, TPQ is an isosceles triangle.

$$\text{Therefore, } \angle TPQ = \angle TQP = \frac{1}{2} (180^\circ - \theta) = 90^\circ - \frac{1}{2} \theta$$

Also, by Theorem 10.1, $\angle OPT = 90^\circ$

$$\begin{aligned} \text{So, } \angle OPQ &= \angle OPT - \angle TPQ = 90^\circ - \left(90^\circ - \frac{1}{2} \theta \right) \\ &= \frac{1}{2} \theta = \frac{1}{2} \angle PTQ \end{aligned}$$

This gives $\angle PTQ = 2 \angle OPQ$

Example 3 : PQ is a chord of length 8 cm of a circle of radius 5 cm. The tangents at P and Q intersect at a point T (see Fig. 10.10). Find the length TP.

Solution : Join OT. Let it intersect PQ at the point R. Then ΔTPQ is isosceles and TO is the angle bisector of $\angle PTQ$. So, $OT \perp PQ$ and therefore, OT bisects PQ which gives $PR = RQ = 4$ cm.

$$\text{Also, } OR = \sqrt{OP^2 - PR^2} = \sqrt{5^2 - 4^2} \text{ cm} = 3 \text{ cm.}$$

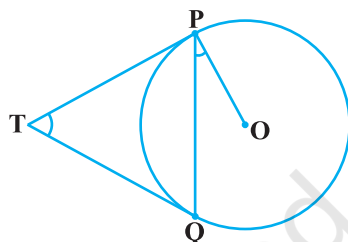


Fig. 10.9

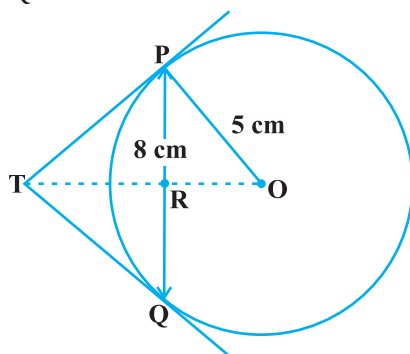


Fig. 10.10